

Acceleration of an Antarctic outlet glacier driven by surface meltwater input to the base

Corresponding Author: Professor Shin Sugiyama

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Review of Sugiyama et al.

The main finding of this paper is of a melt-driven acceleration event on Langhvide Glacier in East Antarctica. As the paper notes, events like these are relatively common on mountain glaciers and the Greenland Ice Sheet but their occurrence in Antarctica is disputed in the literature. This is an important paper as it confirms beyond doubt that melt-driven acceleration events also occur in Antarctica and it does so with a comprehensive dataset from borehole sensors and surface instruments. The paper is very well written, and the figures are of high quality. This paper will make a strong contribution to the literature on this topic and has potential interest to a broad readership. The conclusions of this paper are supported by the data presented but I do wonder whether some of the other key points presented could be emphasised further by inclusion within the abstract and perhaps the introduction and the discussion as this would broaden the paper's reach to a wider audience (see general comment below on this). I also think the paper could be improved by greater reference to the literature on melt-driven acceleration events and atmospheric rivers, and further analysis, for example, of reanalysis data would be warranted. Given the focus of this paper on melt events it would be good to understand the drivers of these processes and also how common they are, as this would put these individual events into their wider context. Finally, there are a few aspects of the GNSS methods that could also be improved especially the description of the interpolation of data gaps.

General Comments

1. In addition to the speed up events the abstract should mention the other key findings of this paper such as the grounding zone conditions and the fauna at the grounding zone. The fact that you can accurately locate the grounding zone is a major strength of this study and it is worth mentioning in the abstract that the area exposed to sea water is larger than that predicted from the hydrostatic pressure (L246). These aspects will be of interest to biologists and those modelling grounding line dynamics respectively.

2. The glacier speed-up on 2-6 January driven by low clouds and warm air temperatures particularly during the night are consistent with the effect of an atmospheric river. Similar events have been described in the literature (e.g. Doyle et al., 2015; Mattingley et al., 2018; Francis et al., 2020; Ing et al., 2024) and it would be worth expanding on this in the discussion. Atmospheric rivers are likely key to these events in Antarctica in the present day and it would be good to touch on this. How often did similar events occur in the AWS and reanalysis data? Are they common, rare or even first-time events?

3. The temperature profile obtained by the thermistor strings as described in the methods is not reported in the results or discussion except for a figure in the supplement. Consider describing these data in the results briefly.

4. I have several questions regarding Figure 3 and the associated methods:

a. How were the missing velocity data interpolated? Interpolation methods were not described in the methods and they look to have been guessed. These are significant gaps in the records at key times so the interpolation needs to be done robustly with a comprehensive description given.

b. Why are some of the shades present/absent on only some of the sub-plots? Why not include all shades on all subplots? If

not, consider explaining why in the caption.

c. The water levels here don't make sense for a ~500-m-thick glacier. A water level of 80 m above the bed does not match a water level of 90% of the ice-overburden pressure. There is something wrong here.

d. In the text water pressure is referred to as percentage of overburden and it would help the reader if a second y-axis were added with pressure on this scale.

e. The floatation fraction will vary with ice thickness so this needs to be given as a range representative of the ice thickness at the different boreholes rather than as a constant.

f. The methods described on L284 suggest static differential carrier-phase positioning was used. See King (2004) for a description of the biases associated with static positioning of a moving antenna and consider whether kinematic positioning would be better. Which ephemeris were used in post-processing? Please state which method of processing was used (e.g. differential carrier-phase).

5. Manuscript structure: the last paragraph of the manuscript before the Methods section currently describes the grounding line conditions, which is a key point but which was absent from the abstract. What is missing here is a conclusion regarding the key story of melt-driven acceleration. There is also little discussion of this process in the discussion section beyond a simple confirmation that it happened. Can you put these events in context of the literature and the long-term record and also describe how widespread they were in the region?

Minor comments

L21 – I agree that the speed up events strongly suggest surface meltwater reached the bed but think you can be even stronger with the language here, that is state that it 'confirms' this.

L48 – can you give a reference for the statement that meltwater penetration to the bed is considered uncommon in Antarctica?

L89 – 'sediment and small gravel'. Small gravel is sediment – just a different grain size so the term sediment needs qualifying here as 'fine-grained sediment' or clay/silt.

L77 – Yes, this points to an 'englacial hydrological network' but this likely takes the form of a basal crevasse which you demonstrate later. I think you should state that this is a basal crevasse here and later discuss this with reference to the literature on basal crevasses (e.g. Harper et al., 2010).

L172 – specify that the ice-bed separation is caused by pressurised subglacial water.

L176 – spelling mistake: Subglacial

L182 – spelling mistake: 'Rutford'.

L202 – it's a bit unclear without reading the references what is meant by 'tidally modulated glacier dynamics'. Can you expand to clarify?

L214 – Consider adding a citation for the statement that subglacial discharge plays a critical role in ice shelf melt and circulation.

L249 – spelling mistake: Whillans

L303 – spelling mistake: Agisoft

L535 – define GNSS acronym on first occurrence and also give acronym for AWS.

L537 – write out "Figures"

Minor Comments on Supporting Information

L31 – label the correlation coefficients on (d) and its inset.

References

Doyle, S. H., Hubbard, A., Van De Wal, R. S., Box, J. E., Van As, D., Scharer, K., ... & Hubbard, B. (2015). Amplified melt and flow of the Greenland ice sheet driven by late-summer cyclonic rainfall. *Nature Geoscience*, 8(8), 647-653.

Harper, J. T., Bradford, J. H., Humphrey, N. F., & Meierbachtol, T. W. (2010). Vertical extension of the subglacial drainage system into basal crevasses. *Nature*, 467(7315), 579-582.

Francis, D., Mattingly, K. S., Temimi, M., Massom, R., & Heil, P. (2020). On the crucial role of atmospheric rivers in the two major Weddell Polynya events in 1973 and 2017 in Antarctica. *Science advances*, 6(46), eabc2695.

Ing, R. N., Nienow, P. W., Sole, A. J., Tedstone, A. J., & Mankoff, K. D. (2024). Minimal impact of late-season melt events on Greenland ice sheet annual motion. *Geophysical Research Letters*, 51(4), e2023GL106520.

King, M. (2004). Rigorous GPS data-processing strategies for glaciological applications. *Journal of Glaciology*, 50(171), 601-607.

Mattingly, K. S., Mote, T. L., & Fettweis, X. (2018). Atmospheric river impacts on Greenland Ice Sheet surface mass balance. *Journal of Geophysical Research: Atmospheres*, 123(16), 8538-8560.

Reviewer #2

(Remarks to the Author)

This is an important and interesting glaciological study that provides the first evidence for surface meltwater reaching the bed of a glacier in continental Antarctica and influencing its dynamics. The data are well documented and presented, and provide conclusive proof that melt events caused speed-up events at the glacier. For readers well-versed in glaciology this is an exciting result, although for a journal with a wider readership the significance is not articulated clearly enough. Some additional work is required to demonstrate how (and if) the results are novel and significant in a broader sense.

In particular, more context is needed to assess the significance of the surface melt events and speed-ups. Most importantly, it is unclear whether the observed surface-to-bed drainage and speed-ups are a new phenomenon resulting from climate warming, or an ongoing process that simply has not been detected before. For example, it is stated (line 113) that rainfall is rare at the study site, with the previous event 4 years before the study period. No such context is provided for the positive air temperature excursions: are there climatic data that might indicate whether these have increased in frequency? Or perhaps there is no evidence of any trend?

If it can be shown that melt events (and by implication, speed-up events) are increasing in frequency, then it might be argued that the study provides a first indication of changing ice sheet response to climate. Modelling studies predict that warming will cause Antarctica to become more “Greenland-like” with major implications for ice dynamics. Does the study show that melt-driven acceleration has recently begun in Antarctica (albeit in a small way), and by implication may become much more important in future? Or is this normal behaviour? If it is normal behaviour, what are the implications for our understanding of ice-sheet dynamics in Antarctica?

In my view, this is a terrific glaciological study that would grace any specialist journal, but for a Nature journal, there must be a stronger case for why it matters. The new observations of sessile fauna in the sub-shelf cavity are also of great interest, but inclusion of this material in the Discussion section means that the paper changes focus at the very point where the wider implications of the glaciological results need to be driven home. So overall, I think there needs to be a stronger sense of what this paper is all about, and why. If glaciological and faunal observations are both significant, should they be presented in different papers for different audiences?

I have no major comments about the content and presentation of the results, and only a few minor comments. In the Abstract and Introduction, it is stated that surface meltwater makes glaciers “flow faster” or “amplifies ice discharge into the ocean”. This is not entirely true. Surface-to-bed drainage can lead to either ice acceleration or deceleration, depending on how the water is stored and transported at the bed. In a poorly-connected drainage system, water inputs can cause an increase in storage and acceleration, while for a well-connected system increasing water flux can enlarge passages, increase drainage efficiency, and cause slow-downs. Both of these processes happen in Greenland, complicating the relationship between surface melt and ice dynamics. I suggest changing the statements in the Abstract and Introduction to read: “glaciers can flow faster” and “it can amplify ice discharge to the ocean”. These themes then need to be revisited in the Discussion, to emphasise that in the case of Langhovde Glacier, meltwater inputs lead to accelerations.

Line 151: here it is mentioned that the ice is “predominantly cold”. Although sub-freezing, the ice is not very cold by Antarctic standards. Ice temperatures down to -6° or so are not unusual in the Greenland Ice Sheet or Arctic glaciers, where surface-to-bed drainage is routinely observed. This is another place where the wider context should be considered – are such ice temperatures typical for coastal regions in East Antarctica?

Reviewer #3

(Remarks to the Author)

Acceleration of an Antarctic outlet glacier driven by surface meltwater input to the base

by Shin Sugiyama et al

Review by Ian C Willis

Overview

This is an excellent and genuinely novel piece of work that represents a step-change in our understanding of Antarctic Ice Sheet glacier dynamics. For the first time, it presents in-situ measurements of short-term velocity variations of an Antarctic Ice Sheet outlet glacier directly linked to fluctuations in subglacial water pressure. The dataset is robust, carefully collected, and meticulously processed. Measurements span two months of the austral summer and include high-resolution (15-minute) GPS observations from two stations positioned above and below the grounding zone of Langhovde Glacier, East Antarctica. These records capture two short-lived speed-up events, each lasting approximately five days and associated with periods of high surface melt or rainfall. Up-glacier of the grounding line, the accelerations coincide with surface uplift, and during the second event a rise in subglacial water pressure from 93–94% to 97% of overburden was recorded. (Water pressure measurements were not available for the first event.)

These observations provide the first direct and compelling evidence that surface meltwater can access the beds of cold Antarctic glaciers, modulating subglacial water pressure and basal sliding. Although this mechanism has been hypothesised previously—and inferred indirectly from satellite-derived seasonal velocity variability—the present dataset from Sugiyama et

al. conclusively demonstrates it. This finding has important implications for ice-sheet mass-balance assessments that rely on the input–output method, which typically assumes steady flow across grounding lines. It also suggests that as climate warming progresses, basal lubrication in cold Antarctic outlet glaciers may become more widespread, with implications for predictive ice flow modelling.

The manuscript is further strengthened by four additional lines of evidence. First, ice-temperature measurements through both grounded and floating ice confirm that the glacier is predominantly cold, with only a thin temperate layer at the base. Second, CTD measurements at the base of grounded ice reveal the presence of freshwater. Third, borehole video logs show deep englacial fractures, providing the necessary pathway for hydrofracture-driven meltwater drainage to the bed. Finally, the borehole imagery also documents marine life—specifically a sea anemone and stalked sponges—at the grounding line.

The work is exceptionally well presented, with a clear and logical structure and concise, precise prose. Figures and captions are also of high quality and fully informative.

For these reasons, I believe the manuscript fully merits publication. I offer only a small number of minor comments and suggestions, primarily relating to the inclusion of additional relevant references and minor clarifications.

Minor comments

Please check how you refer to the GPS stations throughout the paper, including the figures as you use, e.g., GNSS2, GPS2 and G2 interchangeably but it'd be better to refer to them in one way consistently.

Throughout you use the adjective 'subshelf' but I do not think this is widely used. I believe it should be hyphenated 'sub-shelf'

Line 39. In addition to refs 8 10 and 11, could add, e.g., Mair, D., Willis, I., Fischer, U.H., Hubbard, B., Nienow, P. and Hubbard, A., 2003. Hydrological controls on patterns of surface, internal and basal motion during three “spring events”: Haut Glacier d'Arolla, Switzerland. *Journal of Glaciology*, 49(167), pp.555-567.

L39-41. Clarify this previous work was for a valley glacier, not an outlet glacier from the ice sheet.

L46-7. I think this is a little misleading. You could say something like "While interior regions of the Antarctic ice sheet remain too cold for substantial surface melt, many coastal and ice-shelf margin zones do produce significant meltwater during the summer season, including surface lakes, ponds, and slush. Recent data show these melt-water features are widespread around the ice-sheet margin and increasing in some regions". Then to support this you could refer to:

<https://www.nature.com/articles/s41558-025-02363-5>

<https://www.nature.com/articles/s41561-024-01466-6>

L50-51. Here could also refer to:

<https://tc.copernicus.org/articles/16/3907/2022/>

L53. Suggest '...suggested as a possible driver'

L54-55. This is an important point. If you include the Boxall et al paper two sentences up too for completion, then you could also mention ice shelf thickness changes here too. There is also debate about whether some of the seasonal variations in velocity reported by Tuckett et al were below the grounding line not above the grounding line as initially suggested because they used old grounding line data (Rott et al 2020). See also the discussion in Section 5.2 here:

<https://tc.copernicus.org/articles/18/4723/2024/>

Summarising this debate succinctly would be valuable, I think.

L58-9. Surely, we must have a ref to the Siple Coast work here, e.g.:

<https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2020GL091089>

L77. It says 41 m in the Fig caption. I'd say 41 m here too for consistency.

L79. Suggest: "The second borehole was drilled ~5 m from the first hole..."

L90. Gravel is sediment. Could you say "... sediment, which ranged in size from clay to gravel..."?

L95. What was the detection limit of the CTD? Tell us that the water had a salinity less than that. I assume it was freshwater? Perhaps clarify that.

L182-4. Suggest rewording slightly to "At Whillans Ice Stream, ~300 km from the grounding line, subglacial water pressures

measured in > 20 boreholes were consistently very high, typically 98–100% of the overburden pressure for ice that is ~1000 m thick.”

L184. Reference 10 is wrong here as that paper is not about Whillans Ice Stream.

L185. Could start this sentence “Similarly, water pressure beneath...” And note correct spelling of “Rutford.”

L208-9. Could consider “Thus, only pressure changes due to ocean tides, and not ocean water itself, were transmitted...”

L209. You say “...through subglacial channels...” here but you don't have evidence for channels. Could say 'the subglacial drainage system' or 'conduits' to be generic. Or just delete!

L211-212. Suggest “In that case, however, the pressure variations were out of phase with ...”

L218-219. Suggest “...ice motion is controlled by subglacial water pressure...”

L222. Suggest “Meltwater supply to the glacier base implies water is discharged into the sub-shelf...”

L244. Suggest “...ocean also suggests seaward subglacial drainage ...”

L235-236. Suggest “... grounding zone³¹. Our observations confirm that... subsequent upwelling likely contributes to the ocean ...”

L239-241. suggest “ Under the grounding zone of Langhovde Glacier, a thin (< 10 m) seawater layer extended ~1 km upglacier beyond the estimated grounding line (Fig. 1d and Supplementary Fig. 1).

L270-72. Suggest “...assumption. Active seawater inflow to the grounding zone suggests that the thin water layer is subject to oceanic heat transport, which likely drives sub-ice-shelf melting.”

L273. Ubiquitous means ‘everywhere’, so I'd change this word to “common,” “widespread,” or “frequent”.

L287. You say sampling intervals were 1 s, but what does that equate to in terms of depth? It'd be more useful to know that.

L295. You say the video camera was lowered into the boreholes with the CTD profiler. The word ‘with’ here is rather ambiguous. Do you mean it was lowered ‘at the same time’?

L305. Is there a reference for the ocean tide data from Syowa Station which tells us what instrument is used for measurement? If not, can you tell us here what instrument is used?

L306. You say “...GNSS1 and G2...” here but they're labelled GPS1 and GPS2 on Fig 1b.

L310 and L322 GNSS1 => GPS1 ?

L313. Suggest “...side moraine (Fig. 1b, GPS Fix). GNSS...”

L315. “...of 3–5 mm⁵⁵. We...”

L328. Suggest reorder the depths of the BH1804 thermistors to go from shallowest to deepest, as for the other holes.

L337. “Agisoft”

L338. Suggest “...campaign, a basic automatic weather station (Vaisala WXT530) was operated on the glacier at 54 m a.s.l. (Fig. 1b). The single sensor recording air temperature and liquid precipitation every 5 min was installed 2 m...”

Fig 1 caption. What is the significance of black vs red crosses? Is this above and below the grounding line? Is this also the new vs old boreholes as suggested in the text? Can this be explained in the Heading.

It is very hard to see the box indicating the area covered by the drone image. Can this box be given a colour to make it more visible. What is the source of the contour data?

Many of the boreholes are incorrectly labelled in (c).

A pedantic comment (sorry) but you may wish to add the yellow block to the legend for max. clarity. I assume this is 0oC to -2oC.

Fig 3 caption. See my comment re consistency of GPS & GNSS throughout paper, inc. on figures. Can you tell us what the cause of the data gaps was? How have the dashed lines been drawn? I assume just manually rather than a curve fitting algorithm. Explain.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The vast majority of my comments have been satisfactorily addressed and this remains a strong paper relevant to a broad readership. I have three outstanding minor comments:

1. The accuracy of the GNSS processing would be improved using kinematic carrier phase differential positioning. This is explained in King (2004). It is likely that static processing would result in an apparent increase in precision but not accuracy. Whether or not the glacier is moving by stick-slip is irrelevant to the accuracy of GNSS positioning and this is just used by King (2004) to demonstrate the issue when using static positioning of a moving receiver.
2. Its unclear why broadcast ephemeris were used as using a final ephemeris product would also improve positioning accuracy. The use of broadcast ephemeris should be stated in the methods.
3. In my view the end of the discussion is a power position where the key results of this study need to be stated and put into a wider context. This is a place for the key take home message. Hence, the end of the discussion in my view should place, at least slightly, greater emphasis on the wider implications of Antarctic glacier acceleration events. This issue was also raised in the third comment of Reviewer 2.

Points 1 and 2 do not necessarily need to be addressed by revision as the authors describe the methods that they have used sufficiently and any improvement in GNSS accuracy may only make a marginal difference to the results.

Point 3 is an editorial decision.

Reviewer #2

(Remarks to the Author)

The authors have done an excellent job of responding to the first round of reviews. I understand that, in some cases, the reviewers had different views on some aspects of the manuscript and this meant that a degree of compromise had to be made in revising the text. The most important points have been clarified, and the paper is now very strong and well worthy of publication. I very much hope that this paper received the attention it deserves.

I have no further comments or suggestions.

Reviewer #3

(Remarks to the Author)

Thank you to the authors for dealing so extensively with all my initial comments. They have all been dealt with to my satisfaction. It also appears as though all the comments of the other two referees have been dealt with adequately, including the extra work of putting the findings into the longer climatological context and the addition of Supplementary Figure 3.

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Dear Reviewers,

On behalf of the authors, I would like to thank you for the careful evaluation of the manuscript. We have carefully examined the reviewer reports and revised the manuscript for resubmission. In this document, we provide details on our revision and response to the review comments. Reviewers' comments are followed by a reply from the authors in blue letters. The line numbers in our text refer to those in the revised manuscript. In the manuscript, changes from the first version are highlighted in red letters. I believe all the reviewers' comments and suggestions have been addressed.

Best regards,

Shin Sugiyama

Reviewer #1

The main finding of this paper is of a melt-driven acceleration event on Langhvide Glacier in East Antarctica. As the paper notes, events like these are relatively common on mountain glaciers and the Greenland Ice Sheet but their occurrence in Antarctica is disputed in the literature. This is an important paper as it confirms beyond doubt that melt-driven acceleration events also occur in Antarctica and it does so with a comprehensive dataset from borehole sensors and surface instruments. The paper is very well written, and the figures are of high quality. This paper will make a strong contribution to the literature on this topic and has potential interest to a broad readership. The conclusions of this paper are supported by the data presented but I do wonder whether some of the other key points presented could be emphasised further by inclusion within the abstract and perhaps the introduction and the discussion as this would broaden the paper's reach to a wider audience (see general comment below on this). I also think the paper could be improved by greater reference to the literature on melt-driven acceleration events and atmospheric rivers, and further analysis, for example, of reanalysis data would be warranted. Given the focus of this paper on melt events it would be good to understand the drivers of these processes and also how common they are, as this would put these individual events into their wider context. Finally, there are a few aspects of the GNSS methods that could also be improved especially the description of the interpolation of data gaps.

We thank the reviewer for the constructive comments. The manuscript has been revised to address the comments and suggestions. Please see our replies to individual comments as well as the main changes listed below.

- Findings near the grounding line are included in the abstract.
- The occurrence of melt events is analyzed and discussed with additional citations.
- More details are given about GNSS data processing.

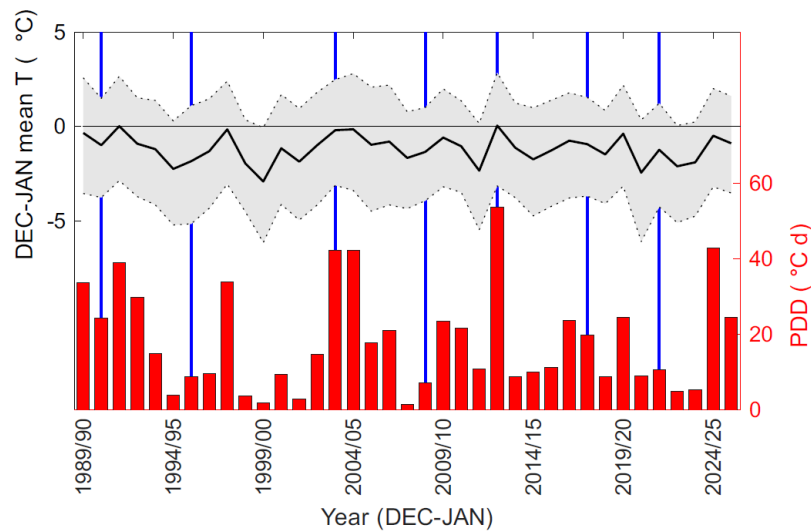
General Comments

1. In addition to the speed up events the abstract should mention the other key findings of this paper such as the grounding zone conditions and the fauna at the grounding zone. The fact that you can accurately locate the grounding zone is a major strength of this study and it is worth mentioning in the abstract that the area exposed to sea water is larger than that predicted from the hydrostatic pressure (L246). These aspects will be of interest to biologists and those modelling grounding line dynamics respectively.

Following the suggestion, the thin seawater layer and sessile animals are described in the abstract (Line 24-27).

2. The glacier speed-up on 2-6 January driven by low clouds and warm air temperatures particularly during the night are consistent with the effect of an atmospheric river. Similar events have been described in the literature (e.g. Doyle et al., 2015; Mattingley et al., 2018; Francis et al., 2020; Ing et al., 2024) and it would be worth expanding on this in the discussion. Atmospheric rivers are likely key to these events in Antarctica in the present day and it would be good to touch on this. How often did similar events occur in the AWS and reanalysis data? Are they common, rare or even first-time events?

Thank you very much for the important suggestion. We analyzed meteorological conditions at Syowa Station from 1989 to 2026 and found that the conditions in the summer of 2021/2022 were not unusual (see the figure below). We can assume similar melt events frequently occur on Langhovde and on nearby glaciers. Rain events are not very common, but they occur every ~5 years. No significant trend is observed in summer temperatures and rain frequency. These points are described in the paragraph added to the first subsection of the Discussion section (Line 187-204). The plot is included as Supplementary Figure 3.



Supplementary Figure 3. Summer (December and January) meteorological conditions at Syowa Station from 1989/20 to 2025/26. Mean air temperature (black) with mean daily minimum/maximum temperatures (dotted), the sum of positive degree days (red), and rain events (blue).

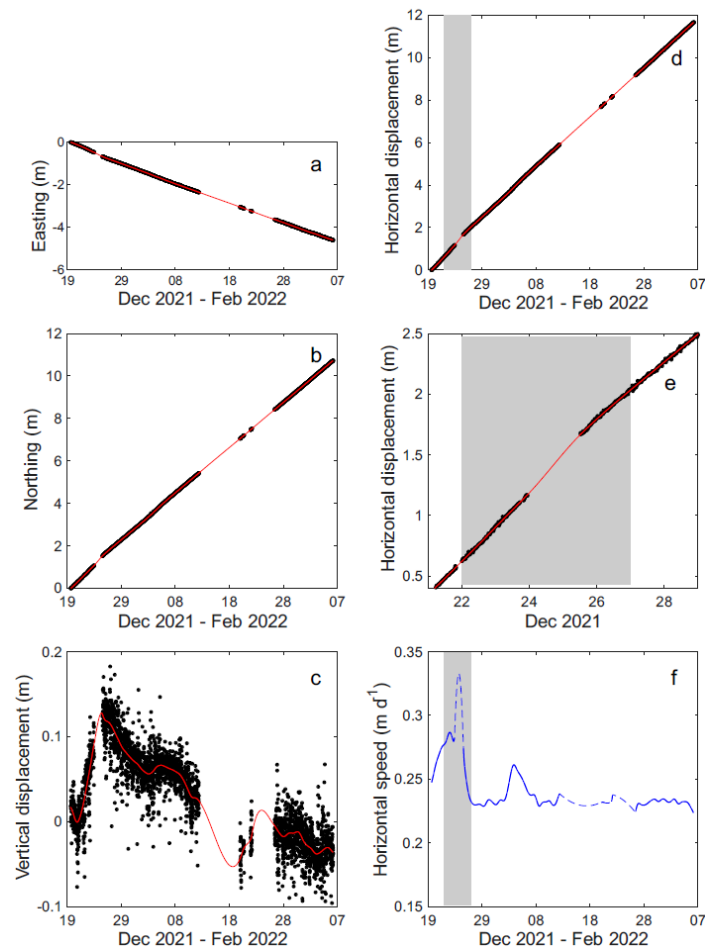
3.The temperature profile obtained by the thermistor strings as described in the methods is not reported in the results or discussion except for a figure in the supplement. Consider describing these data in the results briefly.

Ice temperature data were briefly described in the second paragraph of the Discussion section. We revised the text following a comment from Reviewers #2 and #3 (Line 159-164).

4.I have several questions regarding Figure 3 and the associated methods:

a. How were the missing velocity data interpolated? Interpolation methods were not described in the methods and they look to have been guessed. These are significant gaps in the records at key times so the interpolation needs to be done robustly with a comprehensive description given.

The data gaps in the GNSS measurements were filled using spline interpolation. The procedure is described in the Method section (Line 326-329) with a supplementary figure (Supplementary Fig. 6, see below).



Supplementary Figure 6. Filtering the positioning data and ice speed estimation for the data gaps. (a–c) GNSS positioning data obtained at GNSSI for easting, northing and vertical coordinates (black) and the results of the filtering and spline interpretation (red). (d) Horizontal displacement computed from the easting and northing coordinates before (black) and after the filtering and interpretation (red). (e) Enlarged plot of (d) for the first speed-up (Period I) highlighted with the grey band. (f) Ice speed obtained by the differentiation of the horizontal displacement. The dashed lines correspond to the data gap periods.

b. Why are some of the shades present/absent on only some of the sub-plots? Why not include all shades on all subplots? If not, consider explaining why in the caption.

The shades are provided only for plots that are of interest and referred to in the text. For example, shade is given only in (b) because water pressure variations are of interest during this period. The caption has been revised for clarification (Line 664-665).

c. The water levels here don't make sense for a ~500-m-thick glacier. A water level of 80 m above the bed does not match a water level of 90% of the ice-overburden pressure. There is something wrong here.

The left axis of Figure 3b is in meters above sea level (m a.s.l.). It's not above the bed.

d. In the text water pressure is referred to as percentage of overburden and it would help the reader if a second y-axis were added with pressure on this scale.

This is a good suggestion. The water level relative to the ice overburden in % is given in the light y-axis of Fig. 3b.

e. The floatation fraction will vary with ice thickness so this needs to be given as a range representative of the ice thickness at the different boreholes rather than as a constant.

We drilled two boreholes apart ~5 m, and the depths were very similar (552.0 and 552.5 m). Other boreholes are drilled through floating ice, thus not relevant to the flotation level.

f. The methods described on L284 suggest static differential carrier-phase positioning was used. See King (2004) for a description of the biases associated with static positioning of a moving antenna and consider whether kinematic positioning would be better. Which ephemeris were used in post-processing? Please state which method of processing was used (e.g. differential carrier-phase).

We performed static differential carrier-phase positioning relative to a fixed receiver situated 3–4 km from the glacier sites. Ephemeris transmitted by satellites (Broadcast Ephemeris recorded as a navigation file) was used for the post processing. I wonder if the reviewer is assuming processing with a single GNSS receiver, as discussed in the suggested reference (King, 2004). Because data from a fixed reference receiver is available, static positioning achieves far better accuracy than kinematic positioning. Further, unlike the data discussed by King (2004), our data did not show stick-slip motion, which may require careful treatment of GNSS data. The Method section is updated to clarify the GNSS processing (Line 310-329).

5. Manuscript structure: the last paragraph of the manuscript before the Methods section currently describes the grounding line conditions, which is a key point but which was absent from the abstract. What is missing here is a conclusion regarding the key story of melt-driven acceleration. There is also little discussion of this process in the discussion section beyond a simple confirmation that it happened. Can you put these events in context of the literature and the long-term record and also describe how widespread they were in the region?

An additional analysis of meteorological conditions is described in the first subsection of the Discussion section (Line 186-203). We prefer to place this paragraph in the first subsection (not right before the Method section) for readability.

Minor comments

L21 – I agree that the speed up events strongly suggest surface meltwater reached the bed but think you can be even stronger with the language here, that is state that it ‘confirms’ this.

Corrected as suggested (Line 21).

L48 – can you give a reference for the statement that meltwater penetration to the bed is considered uncommon in Antarctica?

The sentence is revised as “the penetration of surface meltwater to the glacier base has never been confirmed in Antarctica” (Line 54).

L89 – ‘sediment and small gravel’. Small gravel is sediment – just a different grain size so the term sediment needs qualifying here as ‘fine-grained sediment’ or clay/silt.

Following the suggestion from Reviewer #3, "sediment and small gravel" is revised as "sediment, which ranged in size from clay to gravel" (Line 96-97).

L77 – Yes, this points to an ‘englacial hydrological network’ but this likely takes the form of a basal crevasse which you demonstrate later. I think you should state that this is a basal crevasse here and later discuss this with reference to the literature on basal crevasses (e.g. Harper et al., 2010).

Corrected as "... connection of the borehole presumably to a basal crevasse (Line 84)".
The suggested literature is referred to in the second paragraph of Discussion (Line 173, Reference 64).

L172 – specify that the ice-bed separation is caused by pressurised subglacial water.

Revised as "... uplift was due to the separation of ice from the bed by pressurized subglacial water" (Line 183-184).

L176 – spelling mistake: Subglacial

Corrected (Line 206).

L182 – spelling mistake: 'Rutford'.

Corrected (Line 212).

L202 – it's a bit unclear without reading the references what is meant by 'tidally modulated glacier dynamics'. Can you expand to clarify?

Reworded by "tidally modulated glacier flow" (Line 232).

L214 – Consider adding a citation for the statement that subglacial discharge plays a critical role in ice shelf melt and circulation.

This paper is added as a reference (Line 245, Reference 62).

Vaňková, I., Asay-Davis, X., Branecky Begeman, C., Comeau, D., Hager, A., Hoffman, M., et al. (2025). Subglacial discharge effects on Antarctic ice-shelf basal melt and the Southern Ocean in a global, coupled ocean—sea-ice model. *Journal of Geophysical Research: Oceans*, 130, e2025JC022999. <https://doi.org/10.1029/2025JC022999>

L249 – spelling mistake: Whillans

Corrected (Line 279).

L303 – spelling mistake: Agisoft

Corrected (Line 343).

L535 – define GNSS acronym on first occurrence and also give acronym for AWS.

GNSS is already defined in Line 108. Because "AWS" is used only on Figure 1b, it's defined in the figure caption (Line 634).

L537 – write out “Figures”

"Figures" is abbreviated as "Figs" throughout the manuscript, following the journal format.

Minor Comments on Supporting Information

L31 – label the correlation coefficients on (d) and its inset.

Correlation coefficients are indicated on the (d) and its inset (Supplementary Fig. 4d).

Reviewer #2 (Remarks to the Author):

This is an important and interesting glaciological study that provides the first evidence for surface meltwater reaching the bed of a glacier in continental Antarctica and influencing its dynamics. The data are well documented and presented, and provide conclusive proof that melt events caused speed-up events at the glacier. For readers well-versed in glaciology this is an exciting result, although for a journal with a wider readership the significance is not articulated clearly enough. Some additional work is required to demonstrate how (and if) the results are novel and significant in a broader sense.

Thank you very much for providing evaluations and suggestions. Below, we address points raised by the reviewer.

In particular, more context is needed to assess the significance of the surface melt events and speed-ups. Most importantly, it is unclear whether the observed surface-to-bed drainage and speed-ups are a new phenomenon resulting from climate warming, or an ongoing process that simply has not been detected before. For example, it is stated (line 113) that rainfall is rare at the study site, with the previous event 4 years before the study period. No such context is provided for the positive air temperature excursions: are there climatic data that might indicate whether these have increased in frequency? Or perhaps there is no evidence of any trend? If it can be shown that melt events (and by implication, speed-up events) are increasing in frequency, then it might be argued that the study provides a first indication of changing ice sheet response to climate. Modelling studies predict that warming will cause Antarctica to become more “Greenland-like” with major implications for ice dynamics. Does the study show that melt-driven acceleration has recently begun in Antarctica (albeit in a small way), and by implication may become much more important in future? Or is this normal behaviour? If it is normal behaviour, what are the implications for our understanding of ice-sheet dynamics in Antarctica?

We thank for this helpful comment. Following the suggestion from this and another reviewers, meteorological conditions over the last 37 years were analyzed. The results show that our observation in 2021/2022 austral summer season was not unusual. Additional text (Line 187-204) and Supplementary Figure 3 are included. Please also see our reply to the “General Comment 2” from Reviewer #1.

In my view, this is a terrific glaciological study that would grace any specialist journal, but for a Nature journal, there must be a stronger case for why it matters. The new observations of sessile fauna in the sub-shelf cavity are also of great interest, but inclusion of this material in the Discussion section means that the paper changes focus at the very point where the wider implications of the glaciological results need to be driven home. So overall, I think there needs to be a stronger sense of what this paper is all about, and why. If glaciological and faunal observations are both significant, should they be presented in different papers for different audiences?

The most important finding of this study is meltwater-driven glacier acceleration associated with subglacial pressure variations. However, the grounding line conditions and subglacial fauna are also relevant to the main finding, because subglacial water was connected to the thin seawater layer, and the grounding line position affects glacier dynamics. The abstract is revised to include these observations to highlight a link among the borehole observations (Line 24-27). Please also consider "General Comment 1" from Reviewer #1, which acknowledges the importance of the subglacial observations near the grounding line.

I have no major comments about the content and presentation of the results, and only a few minor comments. In the Abstract and Introduction, it is stated that surface meltwater makes glaciers “flow faster” or “amplifies ice discharge into the ocean”. This is not entirely true. Surface-to-bed drainage can lead to either ice acceleration or deceleration, depending on how the water is stored and transported at the bed. In a poorly-connected drainage system, water inputs can cause an increase in storage and acceleration, while for a well-connected system increasing water flux can enlarge passages, increase drainage efficiency, and cause slow-downs. Both of these processes happen in Greenland, complicating the relationship between surface melt and ice dynamics. I suggest changing the statements in the Abstract and Introduction to read: “glaciers can flow faster” and “it can amplify ice discharge to the ocean”. These themes then need to be revisited in the Discussion, to emphasise that in the case of Langhovde Glacier, meltwater inputs lead to accelerations.

I understand the point and correct the phrase in Introduction to “it can amplify ice discharge to the ocean” (Line 53). However, we keep the first sentence in the abstract as it is (Glaciers flow faster when the base is lubricated by meltwater penetrating through

the ice). We believe this sentence holds true because it assumes “the base is lubricated” but not just “surface-to-bed drainage”.

Line 151: here it is mentioned that the ice is “predominantly cold”. Although sub-freezing, the ice is not very cold by Antarctic standards. Ice temperatures down to -6° or so are not unusual in the Greenland Ice Sheet or Arctic glaciers, where surface-to-bed drainage is routinely observed. This is another place where the wider context should be considered – are such ice temperatures typical for coastal regions in East Antarctica?

Ice temperature (-7 – -6 degree C near the surface) was warmer than those reported from large fast flowing outlet glaciers, such as Amery (Craven et al., 2009) and Rutford (Barret et al., 2009). This is likely because relatively slow ice flow speed ($\sim 100 \text{ m a}^{-1}$) allows ice to adjust to the coastal temperature condition (annual mean air temperature is -10 degree C). Ice temperature was warmer than the mean air temperature, which may be due to meltwater refreezing. The text is revised to indicate ice temperatures and to explain the ice is warmer than other Antarctic glaciers (Line 159-164). Further discussion requires more detailed data and analysis, which is beyond the scope of this paper.

Reviewer #3 (Remarks to the Author):

Acceleration of an Antarctic outlet glacier driven by surface meltwater input to the base
by Shin Sugiyama et al

Review by Ian C Willis

Overview

This is an excellent and genuinely novel piece of work that represents a step-change in our understanding of Antarctic Ice Sheet glacier dynamics. For the first time, it presents in-situ measurements of short-term velocity variations of an Antarctic Ice Sheet outlet glacier directly linked to fluctuations in subglacial water pressure. The dataset is robust, carefully collected, and meticulously processed. Measurements span two months of the austral summer and include high-resolution (15-minute) GPS observations from two stations positioned above and below the grounding zone of Langhovde Glacier, East Antarctica. These records capture two short-lived speed-up events, each lasting approximately five days and associated with periods of high surface melt or rainfall. Up-glacier of the grounding line, the accelerations coincide with surface uplift, and during the second event a rise in subglacial water pressure from 93–94% to 97% of overburden was recorded. (Water pressure measurements were not available for the first event.)

These observations provide the first direct and compelling evidence that surface meltwater can access the beds of cold Antarctic glaciers, modulating subglacial water pressure and basal sliding. Although this mechanism has been hypothesised previously—and inferred indirectly from satellite-derived seasonal velocity variability—the present dataset from Sugiyama et al. conclusively demonstrates it. This finding has important implications for ice-sheet mass-balance assessments that rely on the input–output method, which typically assumes steady flow across grounding lines. It also suggests that as climate warming progresses, basal lubrication in cold Antarctic outlet glaciers may become more widespread, with implications for predictive ice flow modelling.

The manuscript is further strengthened by four additional lines of evidence. First, ice-temperature measurements through both grounded and floating ice confirm that the glacier is predominantly cold, with only a thin temperate layer at the base. Second, CTD measurements at the base of grounded ice reveal the presence of freshwater. Third, borehole video logs show deep englacial fractures, providing the necessary pathway for

hydrofracture-driven meltwater drainage to the bed. Finally, the borehole imagery also documents marine life—specifically a sea anemone and stalked sponges—at the grounding line.

The work is exceptionally well presented, with a clear and logical structure and concise, precise prose. Figures and captions are also of high quality and fully informative.

For these reasons, I believe the manuscript fully merits publication. I offer only a small number of minor comments and suggestions, primarily relating to the inclusion of additional relevant references and minor clarifications.

We thank the reviewer for the evaluation and comments based on careful reading. Suggested texts, literature and corrections are very helpful to improve the manuscript. We follow most of the suggestions as details given below.

Minor comments

Please check how you refer to the GPS stations throughout the paper, including the figures as you use, e.g., GNSS2, GPS2 and G2 interchangeably but it'd be better to refer to them in one way consistently.

Corrected to "GNSS1 and GNSS2" to be consistent (text, Figures 1 and 4).

Throughout you use the adjective 'subshelf' but I do not think this is widely used. I believe it should be hyphenated 'sub-shelf'

Corrected as suggested.

Line 39. In addition to refs 8 10 and 11, could add, e.g., Mair, D., Willis, I., Fischer, U.H., Hubbard, B., Nienow, P. and Hubbard, A., 2003. Hydrological controls on patterns of surface, internal and basal motion during three "spring events": Haut Glacier d'Arolla, Switzerland. *Journal of Glaciology*, 49(167), pp.555-567.

The suggested article is included in the reference (Line 42, Reference 11).

L39-41. Clarify this previous work was for a valley glacier, not an outlet glacier from the ice sheet.

Clarified as "... reported only from an ice cap on an island in the Peninsula region "
(Line 44).

L46-7. I think this is a little misleading. You could say something like "While interior regions of the Antarctic ice sheet remain too cold for substantial surface melt, many coastal and ice-shelf margin zones do produce significant meltwater during the summer season, including surface lakes, ponds, and slush. Recent data show these melt-water features are widespread around the ice-sheet margin and increasing in some regions".

Then to support this you could refer to:

<https://www.nature.com/articles/s41558-025-02363-5>

<https://www.nature.com/articles/s41561-024-01466-6>

The manuscript is revised following the suggested text and references (Line 47-51).

L50-51. Here could also refer to: <https://tc.copernicus.org/articles/16/3907/2022/>

The suggested reference is included (Line 57, Reference 22).

L53. Suggest '...suggested as a possible driver'

Corrected as suggested (Line 59).

L54-55. This is an important point. If you include the Boxall et al paper two sentences up too for completion, then you could also mention ice shelf thickness changes here too. There is also debate about whether some of the seasonal variations in velocity reported by Tuckett et al were below the grounding line not above the grounding line as initially suggested because they used old grounding line data (Rott et al 2020). See also the discussion in Section 5.2 here:

<https://tc.copernicus.org/articles/18/4723/2024/>

Summarising this debate succinctly would be valuable, I think.

Following the suggestion, the text is revised to include the influence of ice-shelf thickness change on glacier flow, as well as the importance of grounding-line positions for interpreting ice-speed change (Line 59-62).

L58-9. Surely, we must have a ref to the Siple Coast work here, e.g.:
<https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2020GL091089>

The suggested reference is included (Line 66, Reference 29).

L77. It says 41 m in the Fig caption. I'd say 41 m here too for consistency.

The crack on the borehole wall was measured at 41 m above the bed. However, we wrote "water in the borehole drained at ~40 m above" because we are not sure if water began draining when the drill tip was at 41 m. The initiation of the drainage may be at a depth shallower or deeper than 41 m.

L79. Suggest: "The second borehole was drilled ~5 m from the first hole..."

Corrected as suggested (Line 85).

L90. Gravel is sediment. Could you say "... sediment, which ranged in size from clay to gravel..."?

Revised as suggested (Line 96-97).

L95. What was the detection limit of the CTD? Tell us that the water had a salinity less than that. I assume it was freshwater? Perhaps clarify that.

The accuracy of the conductivity measurement is $7 \times 10^{-3} \text{ m S cm}^{-1}$ as described in the Method section (Line 293), which gives a measure of the detection limit of salinity as ~0.01 PSU. This number is given in the text (Line 102).

L182-4. Suggest rewording slightly to "At Whillans Ice Stream, ~300 km from the grounding line, subglacial water pressures measured in > 20 boreholes were consistently very high, typically 98–100% of the overburden pressure for ice that is ~1000 m thick."

Revised as suggested (Line 208-211).

L184. Reference 10 is wrong here as that paper is not about Whillans Ice Stream.

Reference 10 is removed (Line 211).

L185. Could start this sentence “Similarly, water pressure beneath...” And note correct spelling of “Rutford.”

Revised and corrected as suggested (Line 211-212).

L208-9. Could consider “Thus, only pressure changes due to ocean tides, and not ocean water itself, were transmitted...”

Revised as "Thus, only pressure changes due to ocean tides, and not seawater itself, were transmitted..." (Line 227-228).

L209. You say “...through subglacial channels...” here but you don't have evidence for channels. Could say 'the subglacial drainage system' or 'conduits' to be generic. Or just delete!

"subglacial channels" is replaced by "the subglacial drainage system" (Line 228-229).

L211-212. Suggest “In that case, however, the pressure variations were out of phase with ...”

Revised as suggested (Line 231).

L218-219. Suggest “...ice motion is controlled by subglacial water pressure...”

Corrected as suggested (Line 237).

L222. Suggest “Meltwater supply to the glacier base implies water is discharged into the sub-shelf...”

Corrected as suggested (Line 241).

L244. Suggest “...ocean also suggests seaward subglacial drainage ...”

Corrected as suggested (Line 243).

L235-236. Suggest "... grounding zone³¹. Our observations confirm that... subsequent upwelling likely contributes to the ocean ..."

Revised as suggested (Line 247-248).

L239-241. suggest " Under the grounding zone of Langhovde Glacier, a thin (< 10 m) seawater layer extended ~1 km upglacier beyond the estimated grounding line (Fig. 1d and Supplementary Fig. 1).

Revised as suggested (Line 251-252).

L270-72. Suggest "...assumption. Active seawater inflow to the grounding zone suggests that the thin water layer is subject to oceanic heat transport, which likely drives sub-ice-shelf melting."

Revised as suggested (Line 277-279).

L273. Ubiquitous means 'everywhere', so I'd change this word to "common," "widespread," or "frequent".

Replaced by "widespread" (Line 280).

L287. You say sampling intervals were 1 s, but what does that equate to in terms of depth? It'd be more useful to know that.

According to the descending speed of the device, the sampling intervals in depth were ~0.1–0.5 m in depth. This information is added (Line 294).

L295. You say the video camera was lowered into the boreholes with the CTD profiler. The word 'with' here is rather ambiguous. Do you mean it was lowered 'at the same time'?

Yes. The text is revised as "... lowered into the boreholes at the same time by linking together with the CTD profiler" (295-296).

L305. Is there a reference for the ocean tide data from Syowa Station which tells us what instrument is used for measurement? If not, can you tell us here what instrument is used?

The ocean tide data are based on sea level measurements with pressure sensors at Syowa Station. We used a dataset with 30-sec intervals distributed by the Hydrographic and Oceanographic Department, Japan Coast Guard. This information is added to the Method section with a reference (Aoyama et al., 2016) (Line 306-309, Reference 69).

Aoyama, Y., Doi, K., Ikeda, H., Hayakawa, H. and Shibuya, K. 2016. Five years' gravity observation with the superconducting gravimeter OSG#058 at Syowa Station, East Antarctica: gravitational effects of accumulated snow mass, *Geophysical Journal International*, 205 (2), 1290–1304, <https://doi.org/10.1093/gji/ggw078>

L306. You say "...GNSS1 and G2..." here but they're labelled GPS1 and GPS2 on Fig 1b.

Corrected to GNSS to be consistent (Line 310).

L310 and L322 GNSS1 => GPS1 ?

Leave it as GNSS1 to be consistent (Line 314).

L313. Suggest "...side moraine (Fig. 1b, GPS Fix). GNSS..."

Revised as suggested (Line 317).

L315. "...of 3–5 mm⁵⁵. We..."

The reference is introduced as (ref. 55) because m⁵⁵ looks like "m to the power of 55". We will follow the journal format, anyway.

L328. Suggest reorder the depths of the BH1804 thermistors to go from shallowest to deepest, as for the other holes.

Corrected (Line 334).

L337. “Agisoft”

Corrected (Line 343).

L338. Suggest “...campaign, a basic automatic weather station (Vaisala WXT530) was operated on the glacier at 54 m a.s.l. (Fig. 1b). The single sensor recording air temperature and liquid precipitation every 5 min was installed 2 m...”

Revised following the suggestion. Because WXT530 was operated with other instruments (radiation, ice temperature, sonic ranger), we omit "single" to write "The sensor recording air temperature and liquid precipitation" (Line 344-347).

Fig 1 caption. What is the significance of black vs red crosses? Is this above and below the grounding line? Is this also the new vs old boreholes as suggested in the text? Can this be explained in the Heading.

Red crosses are boreholes drilled in this study, while black crosses are those from previous studies. This information is added to the caption (Line 633).

It is very hard to see the box indicating the area covered by the drone image. Can this box be given a colour to make it more visible. What is the source of the contour data?

Light blue color is given to the box. The surface elevation contours are based on the Pléiades digital elevation model on 17 December 2021. The information is given in the caption (Line 637-638).

Many of the boreholes are incorrectly labelled in (c).

Corrected (Figure 1c).

A pedantic comment (sorry) but you may wish to add the yellow block to the legend for max. clarity. I assume this is 0oC to -2oC.

The color scale legend is revised. The temperature scale is from -7 to -1 degree C (Figure 1d).

Fig 3 caption. See my comment re consistency of GPS & GNSS throughout paper, inc. on figures. Can you tell us what the cause of the data gaps was? How have the dashed lines been drawn? I assume just manually rather than a curve fitting algorithm. Explain.

"GPS" are corrected to "GNSS". The data gaps were due to device failure as described in the Method section (Line 325-328). The gaps were filled by spline interpolation, which is now explained with a supplementary figure (Supplementary Figure 6, Line 325-328, Line 656).

Dear Reviewers,

On behalf of the authors, I would like to thank you for the evaluation of the revised manuscript. We have read through the second-round review comments and editorial requests. In this document, we explain our response to the reviewer comments.

Best regards,

Shin Sugiyama

Reviewer #1

The vast majority of my comments have been satisfactorily addressed and this remains a strong paper relevant to a broad readership. I have three outstanding minor comments:

1. The accuracy of the GNSS processing would be improved using kinematic carrier phase differential positioning. This is explained in King (2004). It is likely that static processing would result in an apparent increase in precision but not accuracy. Whether or not the glacier is moving by stick-slip is irrelevant to the accuracy of GNSS positioning and this is just used by King (2004) to demonstrate the issue when using static positioning of a moving receiver.

Based on my experience and knowledge, I understand the reviewer's comment as follows. In the case of kinematic processing, antenna positions are available every epoch (e.g. 1 sec) with much larger scattering (low precision), as compared to static positioning solutions available sparsely (e.g. 1 hr). By applying statistic processing to the high-frequency kinematic positioning data, there is a possibility to obtain more accurate positions. However, special care and technique are required for processing high-frequency low-precision positioning data. Moreover, as described in King (2004), biases in the static solutions are dependent on ice speed, and relatively slow ice motion (1/4-1/3 of the test case in King, 2004) predicts a smaller bias at Langhovde Glacier. Because our static positioning is doing a good job, we would like to keep our GNSS processing methodology.

2. Its unclear why broadcast ephemeris were used as using a final ephemeris product would also improve positioning accuracy. The use of broadcast ephemeris should be stated in the methods.

I would like to thank the reviewer for this advice. This time, we keep our methodology and state the use of a broadcast ephemeris in the Method section (Line 312).

3. In my view the end of the discussion is a power position where the key results of this study need to be stated and put into a wider context. This is a place for the key take home message. Hence, the end of the discussion in my view should place, at least slightly, greater emphasis on the wider implications of Antarctic glacier acceleration events. This issue was also raised in the third comment of Reviewer 2.

I understand the point. However, the final part of the Discussion is about "Sub-shelf fauna near the grounding line", which is not a right place for the implications of observed glacier acceleration. We believe the best is to discuss the point at the end of the discussion about "Meltwater-driven glacier acceleration" (Line 180-197), which is placed first in the Discussion section as the most important message of the paper.

Points 1 and 2 do not necessarily need to be addressed by revision as the authors describe the methods that they have used sufficiently and any improvement in GNSS accuracy may only make a marginal difference to the results. Point 3 is an editorial decision.

Thank you very much once again for reviewing our manuscript. The paper is greatly improved by careful and critical comments from Reviewer #1.

Reviewer #2 (Remarks to the Author):

The authors have done an excellent job of responding to the first round of reviews. I understand that, in some cases, the reviewers had different views on some aspects of the manuscript and this meant that a degree of compromise had to be made in revising the text. The most important points have been clarified, and the paper is now very strong and well worthy of publication. I very much hope that this paper received the attention it deserves. I have no further comments or suggestions.

Thank you very much for understanding the compromise needed during the revision process. The manuscript has been improved by the constructive suggestions from Reviewer #2 to the previous manuscript.

Reviewer #3 (Remarks to the Author):

Thank you to the authors for dealing so extensively with all my initial comments. They have all been dealt with to my satisfaction. It also appears as though all the comments of the other two referees have been dealt with adequately, including the extra work of putting the findings into the longer climatological context and the addition of Supplementary Figure 3.

We are delighted at the favorable comment. Thank you very much for the suggestions and corrections provided in the first-round review, which helped us to improve the paper.

Shin Sugiyama